UNIT 2 · 8 hours

Steady-State Error and Generalized Error Series



Time Domain Analysis of Control Systems

Control Systems (22EEG53)
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Learning objectives

- Use the final value theorem to find steady-state error
- Explain the limitations of the static error constants
- Derive the generalized error coefficients
- Evaluate the error series for polynomial inputs

Syllabus context

Time domain analysis of control systems: standard test signals, time response of zero, first and second order systems, time domain specifications, type of systems, static error constants and steady-state error, generalized error series.

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Steady-state error

$$E(s) = R(s) \,/\, [1 + G(s)H(s)]$$

error signal

$$e_{ss} = \lim_{s \to 0} s\,E(s)$$

final value theorem

- Valid only if sE(s) has no poles on the jω axis or in the RHP (stable closed loop).
- Depends on both the system type and the input.
- Static error constants cover only step, ramp and parabola.

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Limitations of static constants

- They give only the final value of the error: a constant, zero or infinity.
- They cannot tell how the error varies with time when it is infinite.
- They cannot handle other inputs such as $r(t) = 1 + t + t^2/2$ or sinusoids in a simple way.
- The generalized error series expresses the steady-state error as a function of time.

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Generalized error coefficients

1

Error transfer function

$E(s)/R(s) = 1/(1 + G(s)H(s)) = We(s)$.

2

Expand

$We(s) = C0 + C1\, s + (C2/2!)\, s^2 + (C3/3!)\, s^3 + \dots$

3

Coefficients

$Ci = d^i We(s)/ds^i$ at $s = 0$.

4

Error series

$e_{ss}(t) = C0\, r(t) + C1\, \dot{r}(t) + (C2/2!)\, \ddot{r}(t) + \dots$

C0, C1, C2 are the dynamic error coefficients: C0 = 1/(1 + Kp), C1 = 1/Kv, C2 = 1/Ka for the usual cases.

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Numerical: error series for a type 1 system

Problem G = 10/[s(s + 2)], unity feedback. Find C0, C1, C2 and the steady-state error for r(t) = 1 + t + $t^2/2$.

- We = $s(s + 2)/(s^2 + 2s + 10) = 0.2\ s + 0.06\ s^2 + \ldots$
- C0 = 0; C1 = 0.2 (= 1/Kv); C2/2! = 0.06 ⇒ C2 = 0.12
- $r = 1 + t + t^2/2$; $\dot{r} = 1 + t$; $\ddot{r} = 1$
- e_ss(t) = 0 + 0.2(1 + t) + 0.06 × 1

Answer: e_ss(t) = 0.26 + 0.2 t (grows linearly: parabolic input unfollowed)

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Numerical: final value theorem

Problem Unity-feedback system with $G = 100/[s(s + 10)]$. Find the error for a unit ramp using the final value theorem.

- $E(s) = R/(1 + G) = (1/s^2) \cdot s(s + 10)/(s^2 + 10s + 100) = (s + 10)/[s\,(s^2 + 10s + 100)]$
- $s\,E(s) = (s + 10)/(s^2 + 10s + 100)$
- $ess = \lim_{s \to 0} sE(s) = 10/100$

Answer: ess = 0.1 (= 1/Kv, Kv = 10)

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Effect of disturbances and design

Disturbance error

A load disturbance D(s) adds an error; integral action before the disturbance entry point removes a constant disturbance error.

Increase gain K

Reduces the error but may reduce the damping.

Add integrator (type + 1)

Eliminates a constant error: PI controller, lag compensator (Unit 5).

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Key takeaways

- ess = lim s→0 sR(s)/(1 + GH) for a stable system
- Static constants cover only step/ramp/parabola
- Error series: e_ss = C0 r + C1 ṙ + (C2/2!) r̈ + …
- Higher gain or integration improves accuracy

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Practice questions

Q1. Use the final value theorem for G = 20/[s(s+4)], unit ramp.

Q2. Find C0, C1, C2 for G = 5/[s(s + 1)].

Q3. Why must the closed loop be stable before applying the final value theorem?

Q4. Write the generalized error series.